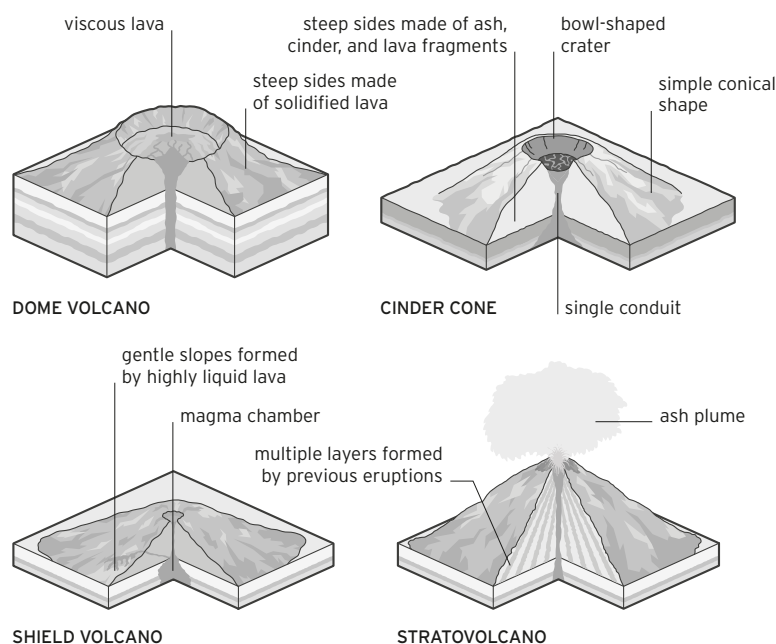


VOLCANO

A volcano is a mountain with an opening, or vent, through which molten lava, volcanic ash, and gases are erupted. Eruptions range from violent andesitic eruptions to the gentle lava flows. Structurally, there are many types of volcanoes, their shapes being largely dependent on the type of eruptions and the quality of the magma. A dome volcano forms steep sides because the viscous lava it erupts cannot flow very far. Stratovolcanoes are formed of many layers of ash and solidified lava.



VOLCANIC PLUG A mass of igneous rock filling the vent of an active volcano. If gas or magma is trapped beneath the plug, pressure will build up, increasing the likelihood of the next eruption being violent. See also *magma*.

VOLCANO See panel, above.

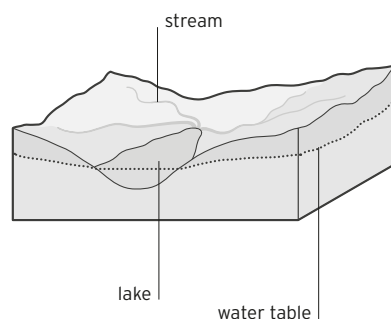
VORTEX A rapidly circulating mass of air or water. See *eddy*, *tornado*, *whirlpool*.

VULCANIAN ERUPTION An explosive type of volcanic eruption. Vulcanian eruptions occur when the pressure of trapped gases in magma becomes strong enough to blow off an overlying crust of solidified magma. See also *eruption* (panel), *magma*.

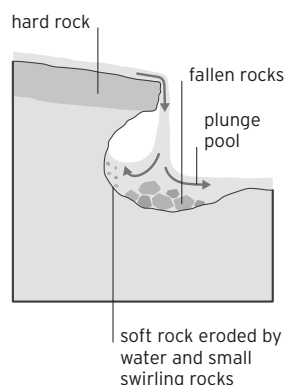
W

WADI A desert riverbed or valley that contains water only at times of heavy rainfall.

WATER TABLE ▼ The upper surface of the groundwater zone. Below the water table, the soil or rocks are permanently saturated, but its position fluctuates, often seasonally. See also *groundwater*.



WATERFALL ▼ A cascade of water formed when a stream or river flows over a cliff or steep incline.



WATERSHED An imaginary line that separates neighboring drainage basins. See also *catchment area*, *drainage basin*.

WAVE A disturbance in the ocean, usually generated by wind. For a wave crossing the sea, the water itself does not move very much, except up and down as the wave passes. The high point of a wave is its crest, and the low point its trough.

WAVELENGTH The distance between successive crests (or troughs) of a wave.

WEATHERING The gradual breakdown of rocks on Earth's surface by chemicals in rainwater, temperature changes, and biological activity such as the growth of lichens and plants. Rocks are broken into small fragments, which are then transported away. See also *erosion*.

WHIRLPOOL A rapidly rotating body of water in a river or the ocean. See also *eddy*, *vortex*.

WHITE SMOKER A hydrothermal vent on the ocean floor emitting light-colored minerals such as those containing barium, calcium, and silicon. White smokers tend to be cooler than black smokers. See also *black smoker*, *hydrothermal vent*.

Y

YARDANG A wind-sculpted desert rock, eroded by windborne particles of sand and dust.

Z

ZOOPLANKTON Animal life, mostly microscopic, that drifts in the water column of oceans, lakes, and rivers. See also *phytoplankton*.

ZOOXANTHELLAE Single-celled algae that live in corals and supply them with vital nutrients to grow. See also *coral bleaching*.